Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Meirinho West, Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
1685 Dunn Road	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 10/01/1989

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X012-X043-X021-XXXX X012-X043-X031-XXXX X012-X058-X003-XXXX X012-X058-X005-XXXX X012-X058-X006-XXXX

B. OPERATORS

Meirinho, Mark

Operator name: Meirinho, Mark

Telephone no.: (209) 578-1403 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5418 Woodland	Modesto	CA	95358

This operator is responsible for paying permit fees.

Meirinho, Tony

Operator name: Meirinho, Tony

Telephone no.: (209) 578-1403 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5418 Woodland	Modesto	CA	95358

C. OWNERS

Meirinho, Mark

Legal owner name: Meirinho, Mark

Telephone no.: (209) 578-1403
Landline | Cellular

5418 Woodland	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Meirinho, Tony

Legal owner name: Meirinho, Tony

Telephone no.: (209) 578-1403
Landline | Cellular

5418 Woodland	Modesto	CA	95358
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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	550	0
Number under roof	2,800	400	1,000	1,000	250	800
Maximum number	3,000	450	1,000	1,000	800	800
Average number	2,800	400	1,000	1,000	800	800
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 103,799.26 *tons per reporting period*

Total nitrogen from manure: 1,292,217.25 *lbs per reporting period*

After ammonia losses (30% loss applied): 904,552.08 *lbs per reporting period*

Total phosphorus from manure: 208,095.55 *lbs per reporting period*

Total potassium from manure: 539,905.03 *lbs per reporting period*

Total salt from manure: 1,410,360.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 28,322,205 *gallons*

Total nitrogen generated: 235,500.66 *lbs*

Total phosphorus generated: 28,043.75 *lbs*

Total potassium generated: 239,112.50 *lbs*

Total salt generated: 1,198,851.94 *lbs*

	28,322,205	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	28,322,205	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
IW #1 Shackleford	Ground water
IW #3 Woodland	Ground water
IW #4 River Road.	Ground water
IW #5 Prune	Ground water

Source Description	Type
IW #6 Greenwood South	Ground water
IW #7 Greenwood NW	Ground water
IW #8 Greenwood NE	Ground water
IW#2 Gates	Ground water
MID Lateral 3	Surface water
River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/12/2023	Solid commercial fertilizer *UN32*	121.05 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	77,394.53	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	77,394.53	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	6	6	2	process wastewater	X012-X058-X006-XXXX
2	18	18	2	process wastewater	X012-X058-X006-XXXX
3	18	18	2	process wastewater	X012-X058-X011-XXXX
4	77	77	2	process wastewater	X012-X058-X011-XXXX
5	49	49	2	process wastewater	X012-X058-X008-XXXX
6	230	230	2	process wastewater	X012-X059-X001-XXXX X012-X059-X008-XXXX X012-X059-X009-XXXX X012-X059-X010-XXXX X012-X059-X011-XXXX X012-X059-X012-XXXX
7	27	27	2	process wastewater	X012-X058-X003-XXXX
Bacon Island	533	533	2	manure	X129-X200-XX42-XXXX
Blue Gum	75	75	2	manure	X012-X035-X002-XXXX X012-X035-X003-XXXX X012-X035-X006-XXXX X012-X035-X007-XXXX
Greenwood	294	294	1	manure	X255-X240-XX02-XXXX X255-X240-XX12-XXXX
Lower Jones	261	261	2	manure	X129-X150-XX07-XXXX
Prune Ranch	63	63	1	manure	X048-X025-X005-XXXX
River Ranch	159	159	1	manure	X016-X010-X010-XXXX X016-X010-X011-XXXX
Willows Ranch	397	397	2	manure	X129-X160-XX01-XXXX
Woodland Ranch	18	18	1	manure	X012-X036-X005-XXXX
Totals for areas that were used for application	2,225	2,225	26		
Totals for areas that were not used for application					
Land application area totals	2,225	2,225	26		

B. CROPS AND HARVESTS

1

Field name: 1

1

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	62.00 *ton*	As-is		53.4	10,200.00	2,430.00	15,300.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	10.33	210.80	50.22	316.20	1,117.16

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 6 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	202.00 *ton*	As-is		72.4	3,230.00	520.00	3,110.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.67	217.49	35.01	209.41	1,263.71

2

Field name: 2

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	233.00 *ton*	As-is		59.6	8,000.00	1,790.00	13,100.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.94	207.11	46.34	339.14	1,087.75

2

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	617.00 *ton*	As-is		72.0	2,940.00	460.00	2,530.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	34.28	201.55	31.54	173.45	1,305.30

3

Field name: 3

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/16/2023	205.00 *ton*	As-is		74.0	12,200.00	1,720.00	23,900.00		14.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	11.39	277.89	39.18	544.39	876.49

06/28/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 18 Plant date: 06/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	226.00 *ton*	As-is		64.0	8,170.00	1,410.00	16,200.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	154.00	23.80	168.00	0.00
Total actual harvest content	12.56	205.16	35.41	406.80	1,166.16

4

Field name: 4

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	1,045.00 *ton*	As-is		62.2	8,500.00	1,850.00	12,600.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.57	230.71	50.21	342.00	1,354.32

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	2,547.00 *ton*	As-is		72.6	3,150.00	480.00	2,660.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	33.08	208.39	31.75	175.97	1,287.00

5

Field name: 5

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	621.00 *ton*	As-is		60.8	8,240.00	1,570.00	14,000.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.67	208.86	39.79	354.86	1,053.22

5

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,603.00 *ton*	As-is		73.8	2,960.00	460.00	2,600.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.71	193.67	30.10	170.11	977.11

6

Field name: 6

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 230 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	3,087.00 *ton*	As-is		69.4	8,050.00	1,550.00	13,500.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.42	216.09	41.61	362.39	1,059.62

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 230 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	7,098.00 *ton*	As-is		71.8	3,500.00	490.00	2,990.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.86	216.03	30.24	184.55	1,148.77

7

Field name: 7

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 27 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	366.00 *ton*	As-is		68.4	9,190.00	1,450.00	12,300.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.56	249.15	39.31	333.47	1,113.72

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 27 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/17/2023	855.00 *ton*	As-is		71.6	2,870.00	480.00	2,780.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.67	181.77	30.40	176.07	1,061.21

Bacon Island

Field name: Bacon Island

11/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 533 Plant date: 11/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	6,751.00 *ton*	As-is		76.6	4,690.00	970.00	2,920.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.67	118.81	24.57	73.97	746.89

Bacon Island

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 533 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	13,576.00 *ton*	As-is		69.0	3,500.00	690.00	3,740.00		21.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.47	178.30	35.15	190.52	3,316.31

Blue Gum

Field name: Blue Gum

11/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 75 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	946.00 *ton*	As-is		61.8	9,080.00	1,740.00	12,900.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.61	229.06	43.89	325.42	1,194.94

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	2,187.00 *ton*	As-is		70.8	3,360.00	500.00	2,960.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.16	195.96	29.16	172.63	1,089.88

Greenwood

Field name: Greenwood

02/01/2017: Almond, in shell

Crop: Almond, in shell Acres planted: 294 Plant date: 02/01/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,754.00 *ton*	As-is		6.2	24,700.00	1,890.00	21,400.00		15.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	301.00	51.10	325.50	0.00
Total actual harvest content	5.97	294.72	22.55	255.34	1,701.21

Lower Jones

Field name: Lower Jones

11/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 261 Plant date: 11/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	3,404.00 *ton*	As-is		77.0	4,830.00	980.00	8,570.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	13.04	125.99	25.56	223.54	665.93

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 261 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	6,578.00 *ton*	As-is		69.4	3,730.00	610.00	3,140.00		15.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.20	188.01	30.75	158.28	2,406.19

Prune Ranch

Field name: Prune Ranch

02/01/2014: Almond, in shell

Crop: Almond, in shell Acres planted: 63 Plant date: 02/01/2014

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	420.00 *ton*	As-is		9.8	17,500.00	1,490.00	21,300.00		18.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	301.00	51.10	325.50	0.00
Total actual harvest content	6.67	233.33	19.87	284.00	2,164.80

River Ranch

Field name: River Ranch

02/01/2007: Almond, in shell

Crop: Almond, in shell Acres planted: 159 Plant date: 02/01/2007

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,046.00 *ton*	As-is		8.4	19,000.00	1,720.00	24,800.00		18.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	301.00	51.10	325.50	0.00
Total actual harvest content	6.58	249.99	22.63	326.30	2,265.78

Willows Ranch

Field name: Willows Ranch

Willows Ranch

11/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 397 Plant date: 11/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	4,963.00 *ton*	As-is		77.4	4,870.00	920.00	7,920.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	12.00	120.00	19.20	99.60	0.00
Total actual harvest content	12.50	121.76	23.00	198.02	638.51

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 397 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	9,520.00 *ton*	As-is		70.4	3,550.00	610.00	3,140.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	23.98	170.26	29.26	150.59	1,320.23

Woodland Ranch

Field name: Woodland Ranch

02/01/2020: Almond, in shell

Crop: Almond, in shell Acres planted: 18 Plant date: 02/01/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	61.00 *ton*	As-is		8.0	21,000.00	2,010.00	21,500.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	3.39	142.33	13.62	145.72	879.21

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/15/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.03	23.84	144.76	657.36	122,445.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	975,000.00 *gal*
Application event totals		103.03	23.84	144.76	657.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	163.73	25.63	130.27	765.97	102,366.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	45.16	877,500.00 *gal*
Application event totals		163.73	25.63	130.27	811.13	

1 - 06/02/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	107.87	5.98	95.69	400.17	62,547.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	55.19	1,072,500.00 *gal*
Application event totals		107.87	5.98	95.69	455.36	

1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.16	877,500.00 *gal*
Application event totals		0.00	0.00	0.00	45.16	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.37	5.50	102.90	533.72	60,148.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	40.14	780,000.00 *gal*
Application event totals		95.37	5.50	102.90	573.86	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	40.14	780,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.14	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.92	5.47	102.41	531.22	59,866.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	40.14	780,000.00 *gal*
Application event totals		94.92	5.47	102.41	571.36	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	40.14	780,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.14	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	40.14	780,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.14	

2 - 11/15/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.64	23.29	141.39	642.08	358,796.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	3,510,000.00 *gal*
Application event totals		100.64	23.29	141.39	642.08	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	169.46	26.52	134.83	792.78	317,845.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	53.52	3,120,000.00 *gal*
Application event totals		169.46	26.52	134.83	846.30	

2 - 06/02/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.35	5.56	89.02	372.28	174,566.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	60.21	3,510,000.00 *gal*
Application event totals		100.35	5.56	89.02	432.49	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	53.52	3,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.52	

2 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.06	5.08	95.02	492.85	166,626.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	51.85	3,022,500.00 *gal*
Application event totals		88.06	5.08	95.02	544.70	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	50.17	2,925,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.17	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.48	5.04	94.38	489.57	165,516.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	53.52	3,120,000.00 *gal*
Application event totals		87.48	5.04	94.38	543.09	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	50.17	2,925,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.17	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	46.83	2,730,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.83	

3 - 11/04/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

3 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.03	22.45	136.32	619.05	345,926.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		97.03	22.45	136.32	619.05	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	176.74	27.66	140.62	826.81	331,491.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	46.83	2,730,000.00 *gal*
Application event totals		176.74	27.66	140.62	873.64	

3 - 06/28/2023: Sudangrass, silage

Field name: 3

Crop: Sudangrass, silage

Plant date: 06/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.14	5.88	94.15	393.74	184,626.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	55.19	3,217,500.00 *gal*
Application event totals		106.14	5.88	94.15	448.93	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	48.50	2,827,500.00 *gal*
Application event totals		0.00	0.00	0.00	48.50	

3 - 06/28/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.53	5.22	97.68	506.65	171,289.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	50.17	2,925,000.00 *gal*
Application event totals		90.53	5.22	97.68	556.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.95	5.01	93.82	486.64	164,526.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	48.50	2,827,500.00 *gal*
Application event totals		86.95	5.01	93.82	535.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	46.83	2,730,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.83	

4 - 11/15/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	108.62	25.13	152.60	693.00	1,656,565.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	11,700,000.00 *gal*
Application event totals		108.62	25.13	152.60	693.00	

4 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	157.37	24.63	125.21	736.20	1,262,629.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	43.79	10,920,000.00 *gal*
Application event totals		157.37	24.63	125.21	779.98	

4 - 06/02/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.18	6.05	96.86	405.04	812,466.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	49.26	12,285,000.00 *gal*
Application event totals		109.18	6.05	96.86	454.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	47.31	11,797,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.24	5.32	99.53	516.25	746,624.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	46.92	11,700,000.00 *gal*
Application event totals		92.24	5.32	99.53	563.16	

4 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	46.92	11,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	89.53	5.16	96.59	501.03	724,612.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	48.48	12,090,000.00 *gal*
Application event totals		89.53	5.16	96.59	549.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	46.92	11,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.35	11,310,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.35	

5 - 11/16/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	95.27	22.05	133.85	607.85	924,656.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	8,580,000.00 *gal*
Application event totals		95.27	22.05	133.85	607.85	

03/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	179.83	28.15	143.09	841.31	918,216.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	52.22	8,287,500.00 *gal*
Application event totals		179.83	28.15	143.09	893.53	

5 - 06/03/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.45	5.57	89.11	372.64	475,661.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	56.52	8,970,000.00 *gal*
Application event totals		100.45	5.57	89.11	429.16	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	53.45	8,482,500.00 *gal*
Application event totals		0.00	0.00	0.00	53.45	

5 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	49.15	7,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.15	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.89	4.72	88.35	458.30	421,788.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	54.07	8,580,000.00 *gal*
Application event totals		81.89	4.72	88.35	512.36	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	51.61	8,190,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.61	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.20	4.39	82.22	426.46	392,486.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	49.15	7,800,000.00 *gal*
Application event totals		76.20	4.39	82.22	475.61	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	49.15	7,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.15	

6 - 11/04/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/04/2022

6 - 11/04/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	102.13	23.63	143.49	651.61	4,652,692.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	35,490,000.00 *gal*
Application event totals		102.13	23.63	143.49	651.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	166.60	26.08	132.55	779.38	3,992,696.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	46.07	34,320,000.00 *gal*
Application event totals		166.60	26.08	132.55	825.45	

6 - 06/03/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.35	6.34	101.44	424.18	2,541,549.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	48.30	35,977,500.00 *gal*
Application event totals		114.35	6.34	101.44	472.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	47.64	35,490,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.64	

6 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	47.91	35,685,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.91	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.75	5.11	95.76	496.69	2,145,693.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	47.64	35,490,000.00 *gal*
Application event totals		88.75	5.11	95.76	544.34	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	47.12	35,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.12	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	88.89	5.12	95.90	497.45	2,148,963.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	46.60	34,710,000.00 *gal*
Application event totals		88.89	5.12	95.90	544.05	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	46.07	34,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.07	

7 - 11/16/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

7 - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.86	22.41	136.08	617.98	517,990.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	4,777,500.00 *gal*
Application event totals		96.86	22.41	136.08	617.98	

03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	178.28	27.91	141.85	834.06	501,592.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	53.52	4,680,000.00 *gal*
Application event totals		178.28	27.91	141.85	887.58	

7 - 06/03/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.63	5.02	80.40	336.23	236,489.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	57.98	5,070,000.00 *gal*
Application event totals		90.63	5.02	80.40	394.21	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	54.63	4,777,500.00 *gal*
Application event totals		0.00	0.00	0.00	54.63	

7 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.41	4.69	87.83	455.59	231,043.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	53.52	4,680,000.00 *gal*
Application event totals		81.41	4.69	87.83	509.11	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	53.52	4,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.52	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.76	4.65	87.14	451.99	229,216.00 *gal*
MID Lateral 3	Surface water	0.00	0.00	0.00	49.06	4,290,000.00 *gal*
Application event totals		80.76	4.65	87.14	501.05	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	46.83	4,095,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.83	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	49.06	4,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.06	

Bacon Island - 11/28/2022: Oats, silage-soft dough

Field name: Bacon Island

Crop: Oats, silage-soft dough Plant date: 11/28/2022

Bacon Island - 11/28/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	155.32	90.13	538.91	7,730.71	4,010.00 *ton*
Application event totals		155.32	90.13	538.91	7,730.71	

03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	99.20	63,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	99.20	

Bacon Island - 06/20/2023: Corn, silage

Field name: Bacon Island

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	203.85	67.82	304.33	4,106.06	3,470.00 *ton*
Application event totals		203.85	67.82	304.33	4,106.06	

05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	108.22	69,120,000.00 *gal*
Application event totals		0.00	0.00	0.00	108.22	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	105.96	67,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	105.96	

Bacon Island - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
River	Surface water	0.00	0.00	0.00	104.56	66,780,000.00 *gal*
Application event totals		32.00	0.00	0.00	104.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	102.58	65,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	102.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	101.17	64,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	101.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	100.05	63,900,000.00 *gal*
Application event totals		0.00	0.00	0.00	100.05	

Blue Gum - 11/17/2022: Oats, silage-soft dough

Field name: Blue Gum

Crop: Oats, silage-soft dough

Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	233.97	135.78	811.82	11,645.54	850.00 *ton*
Application event totals		233.97	135.78	811.82	11,645.54	

Blue Gum - 11/17/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	11,310,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.36	11,017,500.00 *gal*
Application event totals		0.00	0.00	0.00	45.36	

Blue Gum - 06/04/2023: Corn, silage

Field name: Blue Gum

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	229.62	76.40	342.80	4,625.13	550.00 *ton*
Application event totals		229.62	76.40	342.80	4,625.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	48.17	11,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	45.36	11,017,500.00 *gal*
Application event totals		0.00	0.00	0.00	45.36	

Blue Gum - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
MID Lateral 3	Surface water	0.00	0.00	0.00	44.96	10,920,000.00 *gal*
Application event totals		32.00	0.00	0.00	44.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	44.96	10,920,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.35	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	41.75	10,140,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.35	10,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.35	

Greenwood - 02/01/2017: Almond, in shell

Field name: Greenwood

Crop: Almond, in shell

Plant date: 02/01/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Greenwood - 02/01/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	3.72	0.00	0.00	3,141.01	18,629,601.00 *gal*
Application event totals		3.72	0.00	0.00	3,141.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Corral solids	134.12	77.83	465.36	6,675.57	1,910.00 *ton*
Application event totals		134.12	77.83	465.36	6,675.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	7.45	0.00	0.00	6,282.02	37,259,202.00 *gal*
Application event totals		7.45	0.00	0.00	6,282.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	7.45	0.00	0.00	6,282.02	37,259,202.00 *gal*
Application event totals		7.45	0.00	0.00	6,282.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Greenwood - 02/01/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	7.45	0.00	0.00	6,282.02	37,259,202.00 *gal*
Application event totals		7.45	0.00	0.00	6,282.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	7.45	0.00	0.00	6,282.02	37,259,202.00 *gal*
Application event totals		7.45	0.00	0.00	6,282.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	7.45	0.00	0.00	6,282.02	37,259,202.00 *gal*
Application event totals		7.45	0.00	0.00	6,282.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #7 Greenwood NW	Ground water	3.72	0.00	0.00	3,141.01	18,629,601.00 *gal*
Application event totals		3.72	0.00	0.00	3,141.01	

Lower Jones - 11/29/2022: Oats, silage-soft dough

Field name: Lower Jones

Crop: Oats, silage-soft dough

Plant date: 11/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	166.90	96.85	579.09	8,307.00	2,110.00 *ton*
Application event totals		166.90	96.85	579.09	8,307.00	

Lower Jones - 11/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	116.83	36,540,000.00 *gal*
Application event totals		0.00	0.00	0.00	116.83	

Lower Jones - 06/22/2023: Corn, silage

Field name: Lower Jones

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	221.95	73.84	331.34	4,470.48	1,850.00 *ton*
Application event totals		221.95	73.84	331.34	4,470.48	

06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	122.01	38,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	122.01	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	120.28	37,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	120.28	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
River	Surface water	0.00	0.00	0.00	118.56	37,080,000.00 *gal*
Application event totals		32.00	0.00	0.00	118.56	

Lower Jones - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	119.13	37,260,000.00 *gal*
Application event totals		0.00	0.00	0.00	119.13	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	116.25	36,360,000.00 *gal*
Application event totals		0.00	0.00	0.00	116.25	

09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	113.95	35,640,000.00 *gal*
Application event totals		0.00	0.00	0.00	113.95	

Prune Ranch - 02/01/2014: Almond, in shell

Field name: Prune Ranch

Crop: Almond, in shell

Plant date: 02/01/2014

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	261.59	3,989,594.00 *gal*
Application event totals		0.00	0.00	0.00	261.59	

11/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Corral solids	131.08	76.07	454.80	6,524.11	400.00 *ton*
Application event totals		131.08	76.07	454.80	6,524.11	

Prune Ranch - 02/01/2014: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	163.82	7,979,188.00 *gal*
Application event totals		0.00	0.00	0.00	163.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	163.82	7,979,188.00 *gal*
Application event totals		0.00	0.00	0.00	163.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	163.82	7,979,188.00 *gal*
Application event totals		0.00	0.00	0.00	163.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	163.82	7,979,188.00 *gal*
Application event totals		0.00	0.00	0.00	163.82	

Prune Ranch - 02/01/2014: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	163.82	7,979,188.00 *gal*
Application event totals		0.00	0.00	0.00	163.82	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	81.91	3,989,594.00 *gal*
Application event totals		0.00	0.00	0.00	81.91	

River Ranch - 02/01/2007: Almond, in shell

Field name: River Ranch

Crop: Almond, in shell

Plant date: 02/01/2007

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	8.55	0.00	0.00	578.73	10,116,269.00 *gal*
Application event totals		8.55	0.00	0.00	578.73	

11/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Corral solids	132.44	76.85	459.52	6,591.82	1,020.00 *ton*
Application event totals		132.44	76.85	459.52	6,591.82	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	17.10	0.00	0.00	1,157.46	20,232,538.00 *gal*
Application event totals		17.10	0.00	0.00	1,157.46	

River Ranch - 02/01/2007: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	17.10	0.00	0.00	1,157.46	20,232,538.00 *gal*
Application event totals		17.10	0.00	0.00	1,157.46	

05/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	17.10	0.00	0.00	1,157.46	20,232,538.00 *gal*
Application event totals		17.10	0.00	0.00	1,157.46	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	17.10	0.00	0.00	1,157.46	20,232,538.00 *gal*
Application event totals		17.10	0.00	0.00	1,157.46	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	17.10	0.00	0.00	1,157.46	20,232,538.00 *gal*
Application event totals		17.10	0.00	0.00	1,157.46	

River Ranch - 02/01/2007: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW #4 River Road.	Ground water	8.55	0.00	0.00	578.73	10,116,269.00 *gal*
Application event totals		8.55	0.00	0.00	578.73	

Willows Ranch - 11/30/2022: Oats, silage-soft dough

Field name: Willows Ranch

Crop: Oats, silage-soft dough

Plant date: 11/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	163.81	95.06	568.36	8,153.09	3,150.00 *ton*
Application event totals		163.81	95.06	568.36	8,153.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	121.83	57,960,000.00 *gal*
Application event totals		0.00	0.00	0.00	121.83	

Willows Ranch - 06/24/2023: Corn, silage

Field name: Willows Ranch

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Willows Ranch - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	206.65	68.75	308.50	4,162.30	2,620.00 *ton*
Application event totals		206.65	68.75	308.50	4,162.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	125.24	59,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	125.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	123.35	58,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	123.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
River	Surface water	0.00	0.00	0.00	121.45	57,780,000.00 *gal*
Application event totals		32.00	0.00	0.00	121.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	122.21	58,140,000.00 *gal*
Application event totals		0.00	0.00	0.00	122.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	119.94	57,060,000.00 *gal*
Application event totals		0.00	0.00	0.00	119.94	

Willows Ranch - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River	Surface water	0.00	0.00	0.00	118.05	56,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	118.05	

Woodland Ranch - 02/01/2020: Almond, in shell

Field name: Woodland Ranch

Crop: Almond, in shell

Plant date: 02/01/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	0.00	1,254,992.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Corral solids	126.16	73.21	437.75	6,279.46	110.00 *ton*
Application event totals		126.16	73.21	437.75	6,279.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.06	2,509,984.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Woodland Ranch - 02/01/2020: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.06	2,509,984.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.06	2,509,984.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.06	2,509,984.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.06	2,509,984.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

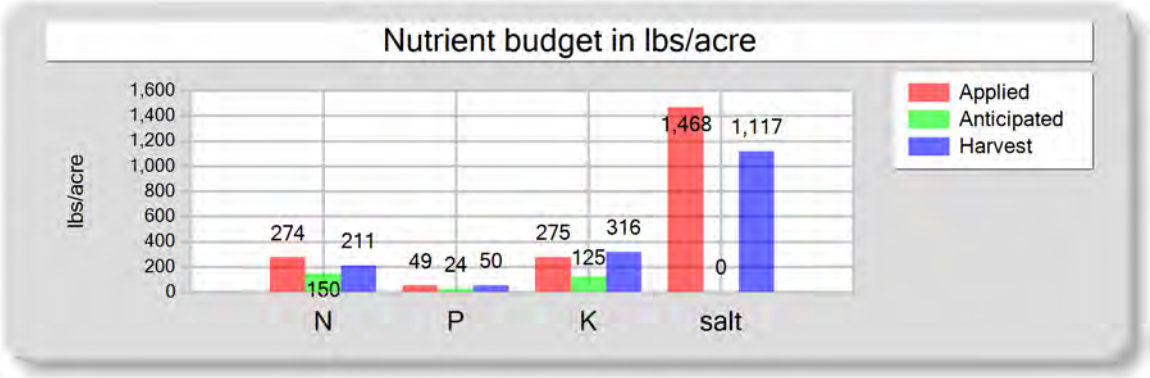
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	21.53	1,254,992.00 *gal*
Application event totals		0.00	0.00	0.00	21.53	

B. NUTRIENT BUDGET

1 - 11/15/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	266.76	49.47	275.03	1,423.33
Fresh water	0.00	0.00	0.00	45.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	273.76	49.47	275.03	1,468.49
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	210.80	50.22	316.20	1,117.16
Nutrient balance	62.96	-0.75	-41.17	351.33
Applied to removed ratio	1.30	0.99	0.87	1.31

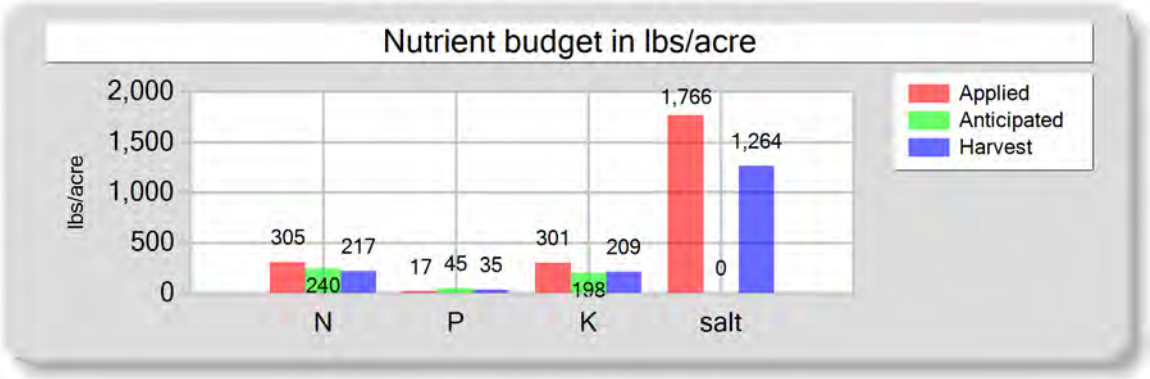
Fresh water applied	
1,852,500.00	*gallons*
68.22	*acre-inches*
11.37	*inches/acre*

Process wastewater applied	
224,811.00	*gallons*
8.28	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/02/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	298.16	16.94	301.00	1,465.11
Fresh water	0.00	0.00	0.00	301.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.16	16.94	301.00	1,766.16
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.49	35.01	209.41	1,263.71
Nutrient balance	87.67	-18.07	91.60	502.45
Applied to removed ratio	1.40	0.48	1.44	1.40

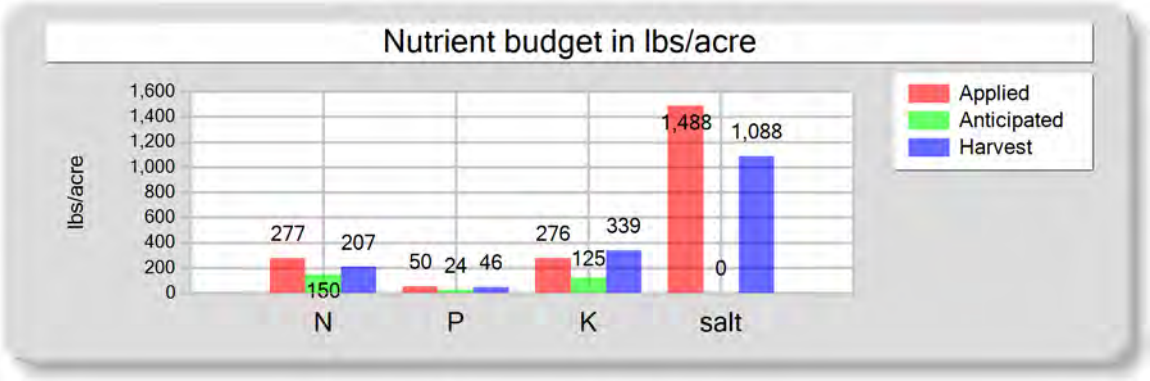
Fresh water applied	
5,850,000.00	*gallons*
215.44	*acre-inches*
35.91	*inches/acre*

Process wastewater applied	
182,561.00	*gallons*
6.72	*acre-inches*
1.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/15/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	270.10	49.81	276.22	1,434.86
Fresh water	0.00	0.00	0.00	53.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.10	49.81	276.22	1,488.38
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	207.11	46.34	339.14	1,087.75
Nutrient balance	69.99	3.47	-62.92	400.63
Applied to removed ratio	1.34	1.07	0.81	1.37

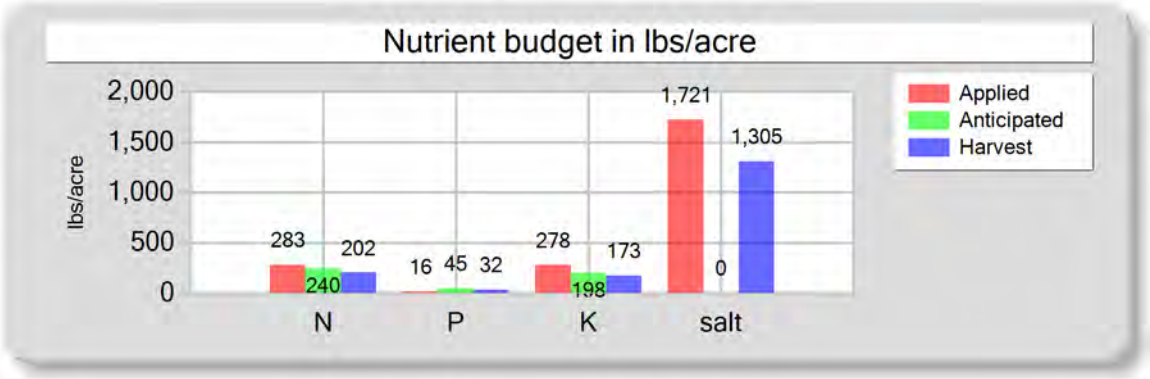
Fresh water applied
6,630,000.00 *gallons*
244.16 *acre-inches*
13.56 *inches/acre*

Process wastewater applied
676,641.00 *gallons*
24.92 *acre-inches*
1.38 *inches/acre*

Total harvests for the crop
1 *harvests*

2 - 06/02/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.90	15.68	278.43	1,354.70
Fresh water	0.00	0.00	0.00	366.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.90	15.68	278.43	1,720.98
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.55	31.54	173.45	1,305.30
Nutrient balance	81.34	-15.86	104.98	415.68
Applied to removed ratio	1.40	0.50	1.61	1.32

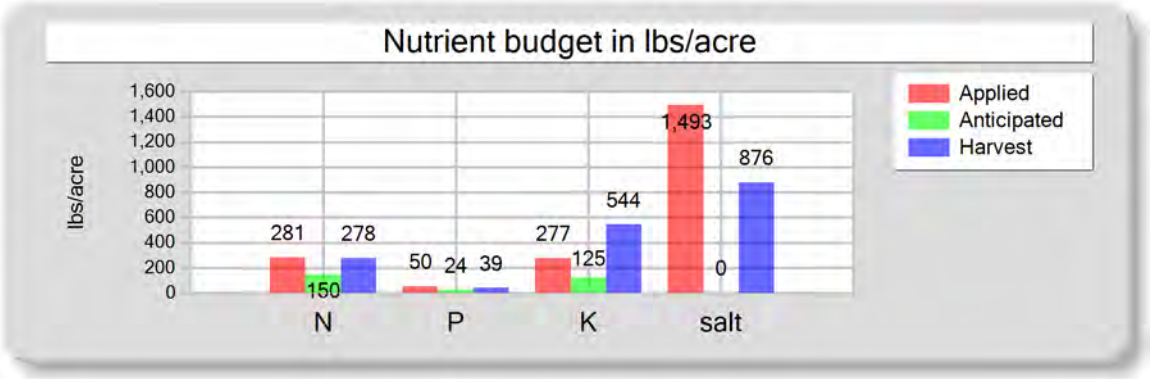
Fresh water applied	
21,352,500.00	*gallons*
786.34	*acre-inches*
43.69	*inches/acre*

Process wastewater applied	
506,708.00	*gallons*
18.66	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/04/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	273.76	50.12	276.94	1,445.86
Fresh water	0.00	0.00	0.00	46.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.76	50.12	276.94	1,492.69
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	277.89	39.18	544.39	876.49
Nutrient balance	2.87	10.94	-267.45	616.20
Applied to removed ratio	1.01	1.28	0.51	1.70

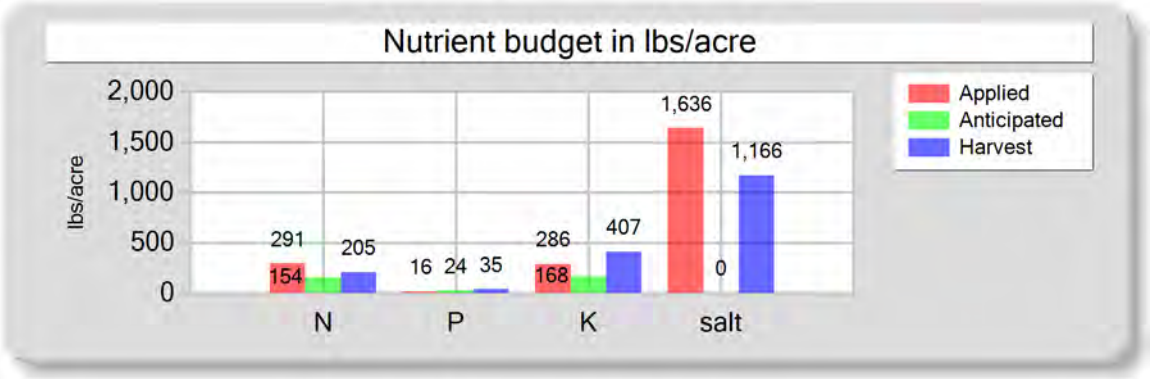
Fresh water applied	
5,850,000.00	*gallons*
215.44	*acre-inches*
11.97	*inches/acre*

Process wastewater applied	
677,417.00	*gallons*
24.95	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/28/2023: Sudangrass, silage

Field name: 3 Crop: Sudangrass, silage Plant date: 06/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	283.62	16.11	285.65	1,387.02
Fresh water	0.00	0.00	0.00	249.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.62	16.11	285.65	1,636.22
Anticipated crop nutrient removal	154.00	23.80	168.00	0.00
Actual crop nutrient removal	205.16	35.41	406.80	1,166.16
Nutrient balance	85.46	-19.30	-121.15	470.06
Applied to removed ratio	1.42	0.45	0.70	1.40

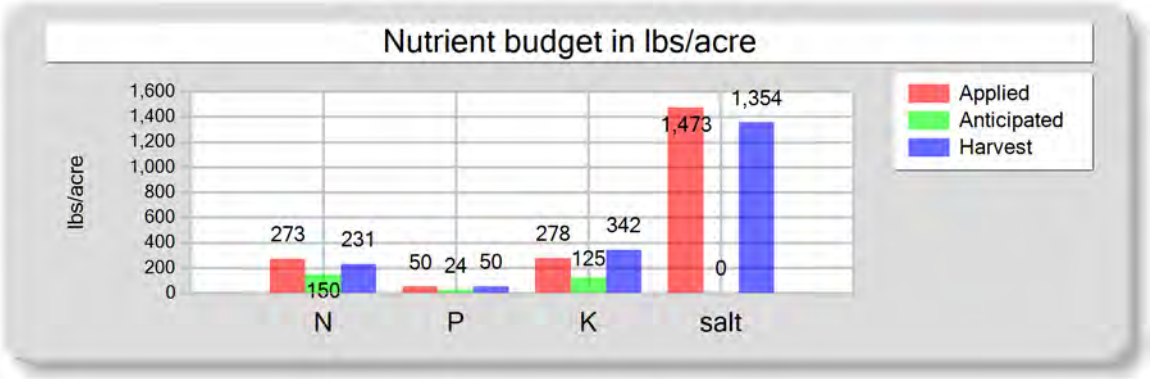
Fresh water applied	
14,527,500.00	*gallons*
535.00	*acre-inches*
29.72	*inches/acre*

Process wastewater applied	
520,441.00	*gallons*
19.17	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 11/15/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.98	49.77	277.81	1,429.19
Fresh water	0.00	0.00	0.00	43.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	272.98	49.77	277.81	1,472.98
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	230.71	50.21	342.00	1,354.32
Nutrient balance	42.27	-0.45	-64.19	118.66
Applied to removed ratio	1.18	0.99	0.81	1.09

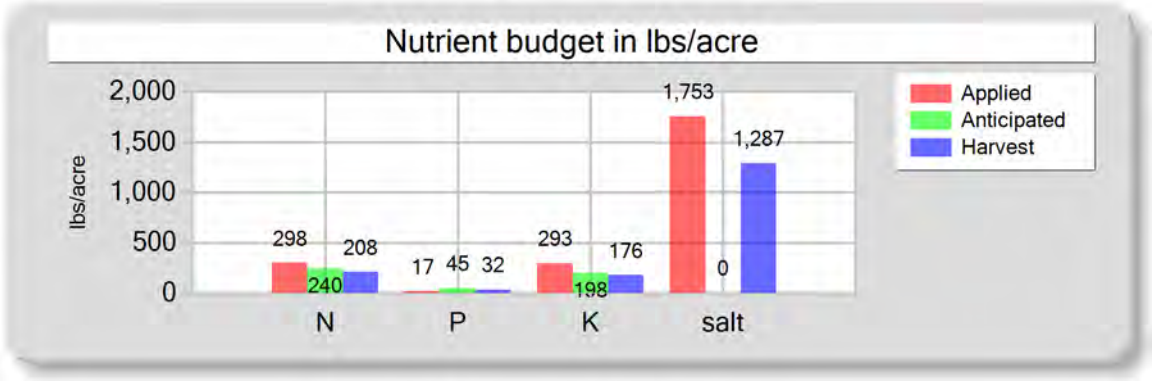
Fresh water applied	
22,620,000.00	*gallons*
833.02	*acre-inches*
10.82	*inches/acre*

Process wastewater applied	
2,919,194.00	*gallons*
107.50	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/02/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.96	16.52	292.98	1,422.32
Fresh water	0.00	0.00	0.00	331.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.96	16.52	292.98	1,753.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	208.39	31.75	175.97	1,287.00
Nutrient balance	89.56	-15.23	117.00	466.47
Applied to removed ratio	1.43	0.52	1.66	1.36

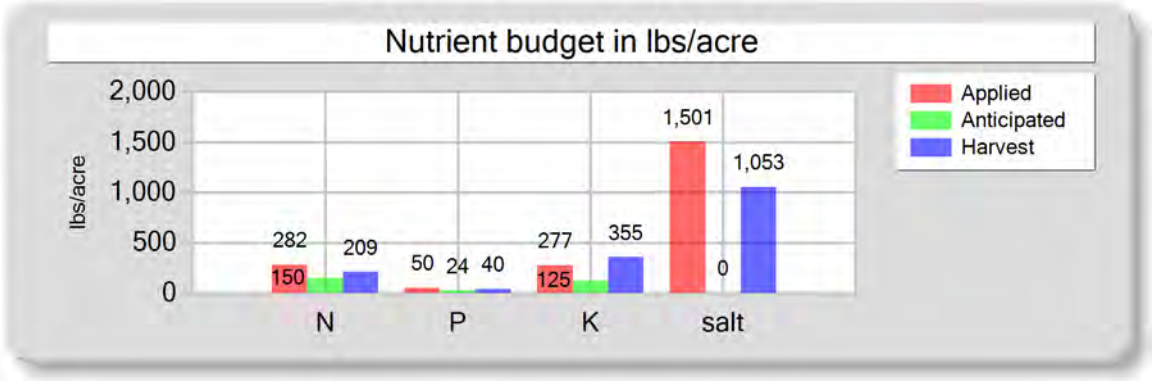
Fresh water applied	
82,582,500.00	*gallons*
3,041.23	*acre-inches*
39.50	*inches/acre*

Process wastewater applied	
2,283,702.00	*gallons*
84.10	*acre-inches*
1.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/16/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.11	50.19	276.94	1,449.16
Fresh water	0.00	0.00	0.00	52.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.11	50.19	276.94	1,501.39
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	208.86	39.79	354.86	1,053.22
Nutrient balance	73.25	10.40	-77.92	448.17
Applied to removed ratio	1.35	1.26	0.78	1.43

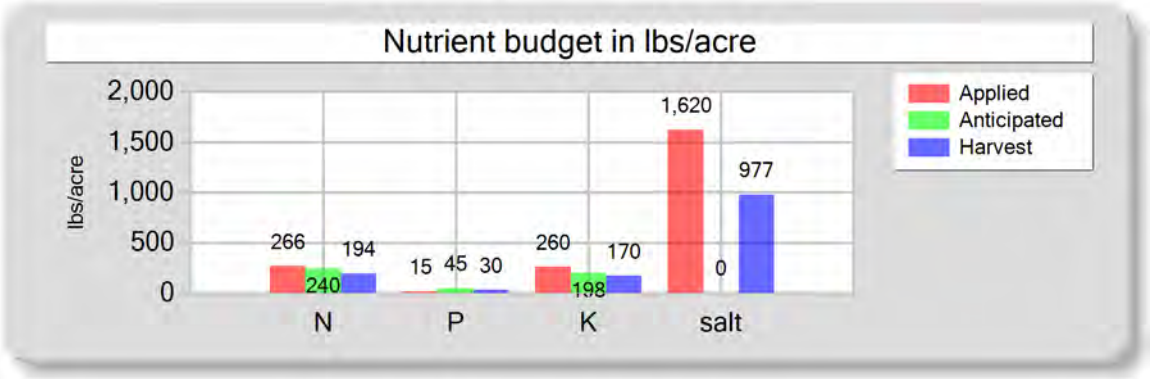
Fresh water applied	
16,867,500.00	*gallons*
621.17	*acre-inches*
12.68	*inches/acre*

Process wastewater applied	
1,842,872.00	*gallons*
67.87	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/03/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.54	14.68	259.68	1,257.39
Fresh water	0.00	0.00	0.00	363.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.54	14.68	259.68	1,620.49
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	193.67	30.10	170.11	977.11
Nutrient balance	71.87	-15.42	89.57	643.38
Applied to removed ratio	1.37	0.49	1.53	1.66

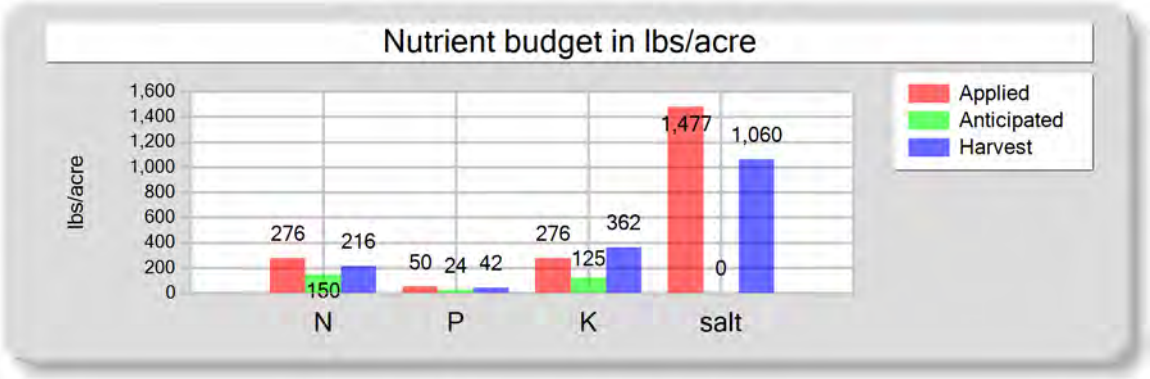
Fresh water applied	
57,622,500.00	*gallons*
2,122.04	*acre-inches*
43.31	*inches/acre*

Process wastewater applied	
1,289,935.00	*gallons*
47.50	*acre-inches*
0.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/04/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	268.73	49.71	276.04	1,430.99
Fresh water	0.00	0.00	0.00	46.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.73	49.71	276.04	1,477.06
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	216.09	41.61	362.39	1,059.62
Nutrient balance	59.64	8.10	-86.35	417.44
Applied to removed ratio	1.28	1.19	0.76	1.39

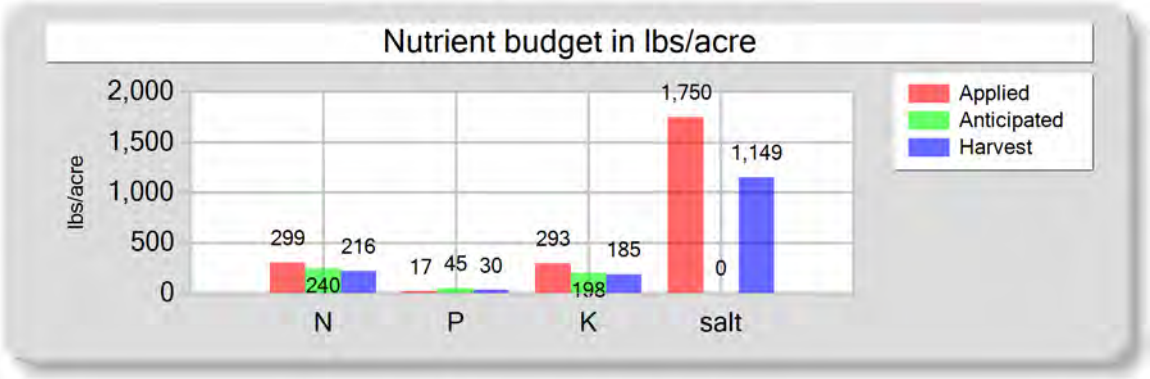
Fresh water applied	
69,810,000.00	*gallons*
2,570.86	*acre-inches*
11.18	*inches/acre*

Process wastewater applied	
8,645,388.00	*gallons*
318.38	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/03/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.98	16.57	293.10	1,418.32
Fresh water	0.00	0.00	0.00	331.28
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.98	16.57	293.10	1,749.61
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	216.03	30.24	184.55	1,148.77
Nutrient balance	82.96	-13.67	108.55	600.84
Applied to removed ratio	1.38	0.55	1.59	1.52

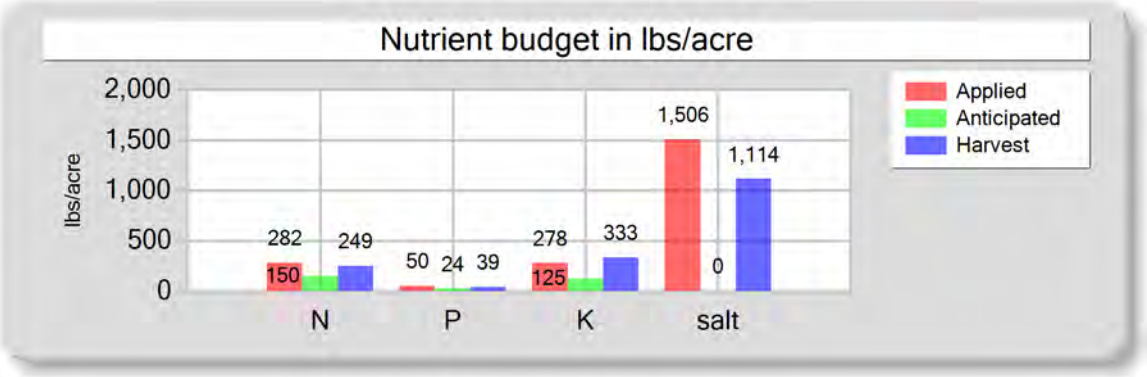
Fresh water applied	
246,772,500.00	*gallons*
9,087.79	*acre-inches*
39.51	*inches/acre*

Process wastewater applied	
6,836,205.00	*gallons*
251.75	*acre-inches*
1.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/16/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.14	50.32	277.93	1,452.03
Fresh water	0.00	0.00	0.00	53.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.14	50.32	277.93	1,505.55
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	249.15	39.31	333.47	1,113.72
Nutrient balance	32.99	11.01	-55.53	391.83
Applied to removed ratio	1.13	1.28	0.83	1.35

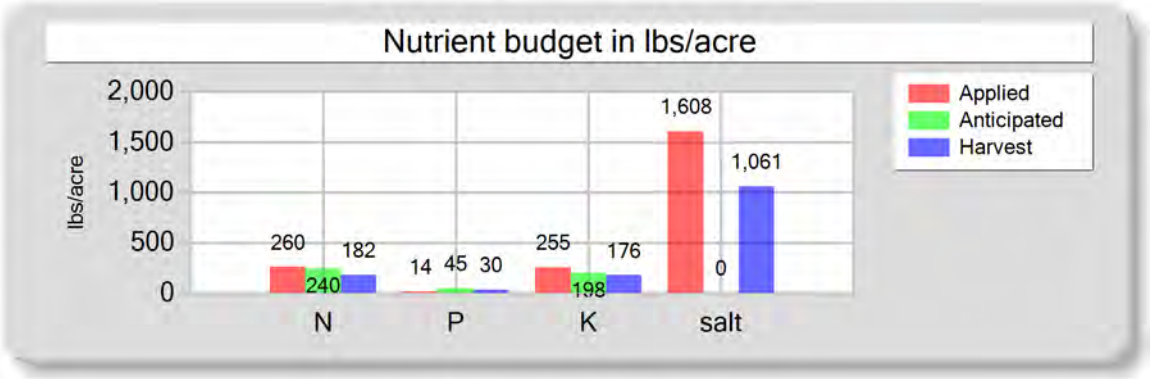
Fresh water applied	
9,457,500.00	*gallons*
348.29	*acre-inches*
12.90	*inches/acre*

Process wastewater applied	
1,019,582.00	*gallons*
37.55	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/03/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.80	14.37	255.37	1,243.81
Fresh water	0.00	0.00	0.00	364.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.80	14.37	255.37	1,608.41
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	181.77	30.40	176.07	1,061.21
Nutrient balance	78.04	-16.03	79.31	547.19
Applied to removed ratio	1.43	0.47	1.45	1.52

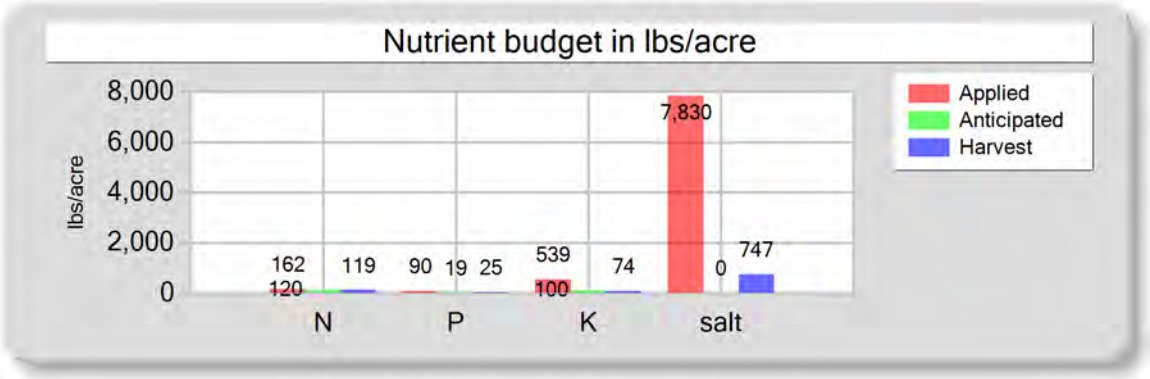
Fresh water applied	
31,882,500.00	*gallons*
1,174.12	*acre-inches*
43.49	*inches/acre*

Process wastewater applied	
696,748.00	*gallons*
25.66	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bacon Island - 11/28/2022: Oats, silage-soft dough

Field name: Bacon Island | Crop: Oats, silage-soft dough | Plant date: 11/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	155.32	90.13	538.91	7,730.71
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	99.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	162.32	90.13	538.91	7,829.91
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	118.81	24.57	73.97	746.89
Nutrient balance	43.51	65.56	464.94	7,083.02
Applied to removed ratio	1.37	3.67	7.29	10.48

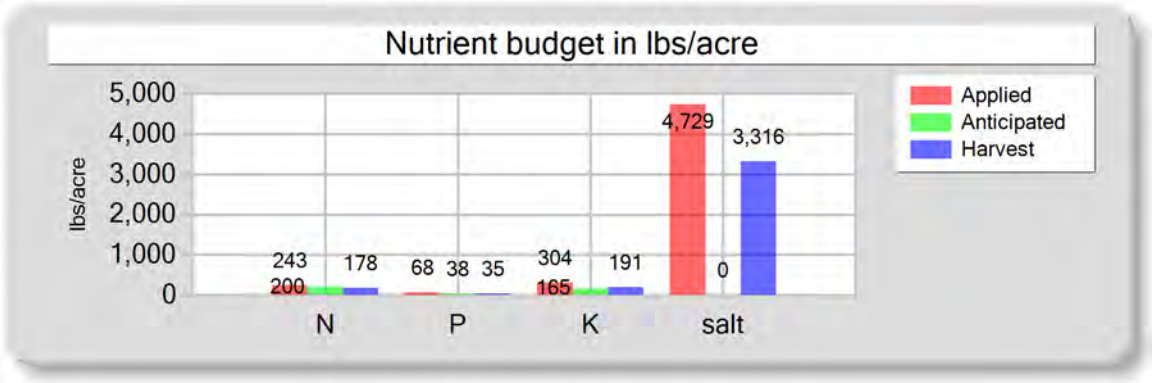
Fresh water applied	
63,360,000.00	*gallons*
2,333.33	*acre-inches*
4.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Bacon Island - 06/20/2023: Corn, silage

Field name: Bacon Island Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	203.85	67.82	304.33	4,106.06
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	622.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.85	67.82	304.33	4,728.60
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	178.30	35.15	190.52	3,316.31
Nutrient balance	64.56	32.67	113.81	1,412.28
Applied to removed ratio	1.36	1.93	1.60	1.43

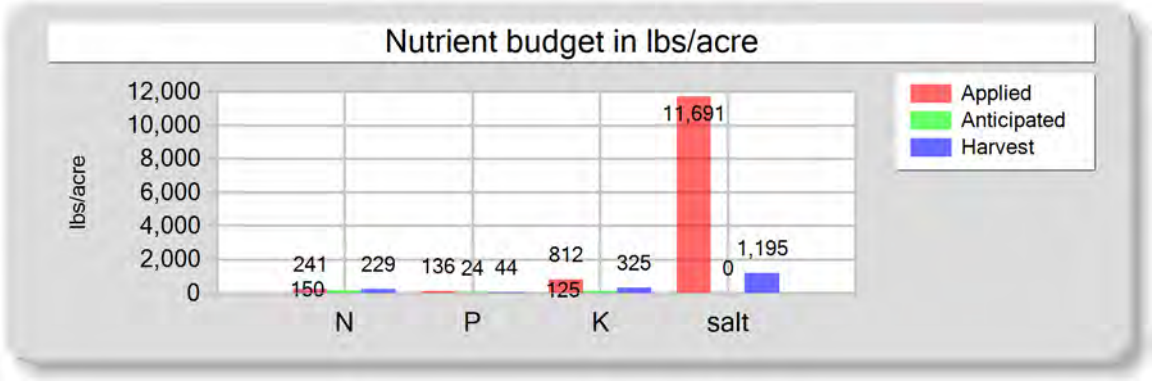
Fresh water applied	
397,620,000.00	*gallons*
14,642.99	*acre-inches*
27.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Blue Gum - 11/17/2022: Oats, silage-soft dough

Field name: Blue Gum Crop: Oats, silage-soft dough Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.97	135.78	811.82	11,645.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	45.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.97	135.78	811.82	11,690.90
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	229.06	43.89	325.42	1,194.94
Nutrient balance	11.92	91.88	486.40	10,495.96
Applied to removed ratio	1.05	3.09	2.49	9.78

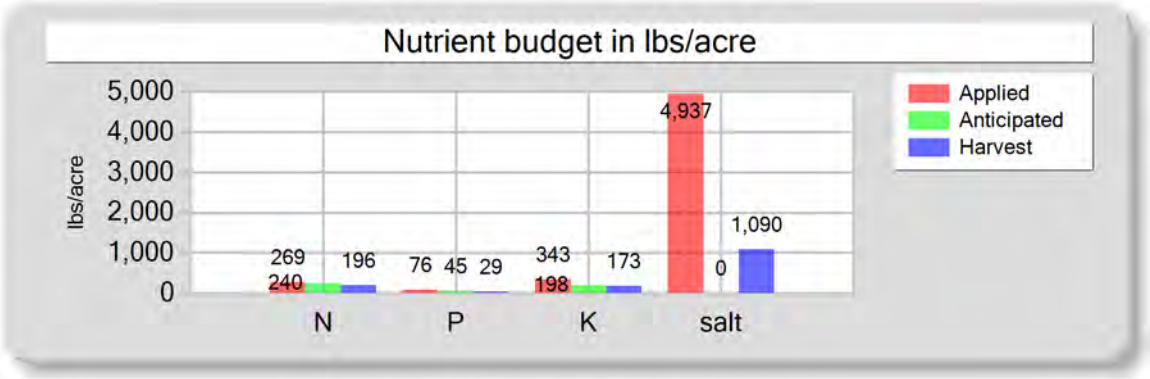
Fresh water applied	
22,327,500.00	*gallons*
822.25	*acre-inches*
10.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Blue Gum - 06/04/2023: Corn, silage

Field name: Blue Gum Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	229.62	76.40	342.80	4,625.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	311.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	268.62	76.40	342.80	4,937.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	195.96	29.16	172.63	1,089.88
Nutrient balance	72.67	47.24	170.18	3,847.13
Applied to removed ratio	1.37	2.62	1.99	4.53

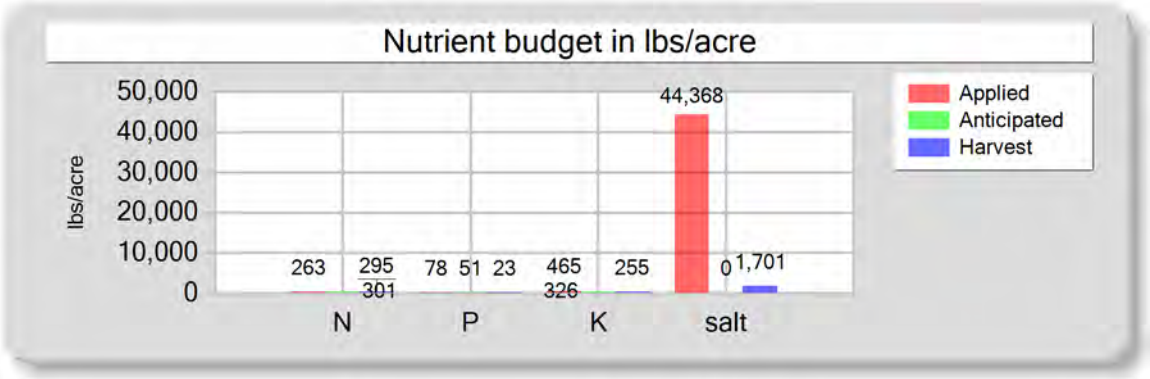
Fresh water applied	
75,757,500.00	*gallons*
2,789.89	*acre-inches*
37.20	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Greenwood - 02/01/2017: Almond, in shell

Field name: Greenwood Crop: Almond, in shell Plant date: 02/01/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	134.12	77.83	465.36	6,675.57
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	44.67	0.00	0.00	37,692.09
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	263.19	77.83	465.36	44,367.66
Anticipated crop nutrient removal	301.00	51.10	325.50	0.00
Actual crop nutrient removal	294.72	22.55	255.34	1,701.21
Nutrient balance	-31.53	55.28	210.01	42,666.45
Applied to removed ratio	0.89	3.45	1.82	26.08

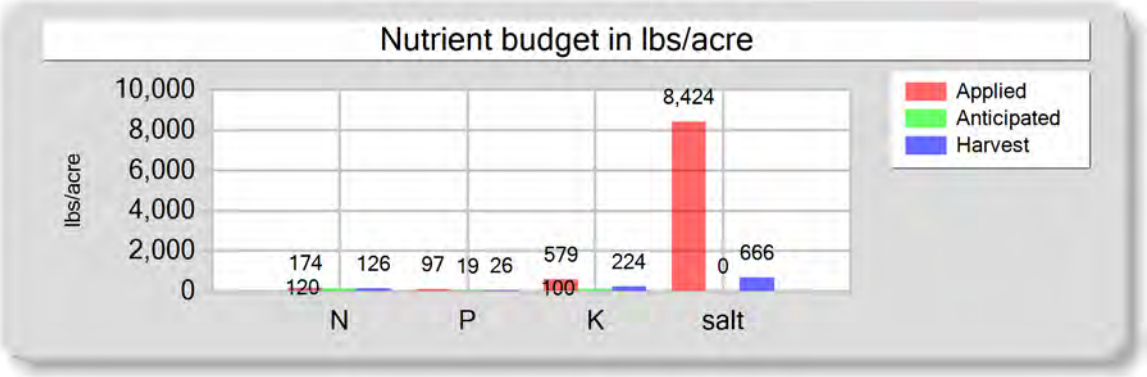
Fresh water applied	
223,555,212.00	*gallons*
8,232.78	*acre-inches*
28.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lower Jones - 11/29/2022: Oats, silage-soft dough

Field name: Lower Jones Crop: Oats, silage-soft dough Plant date: 11/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.90	96.85	579.09	8,307.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	116.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.90	96.85	579.09	8,423.83
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	125.99	25.56	223.54	665.93
Nutrient balance	47.91	71.29	355.55	7,757.90
Applied to removed ratio	1.38	3.79	2.59	12.65

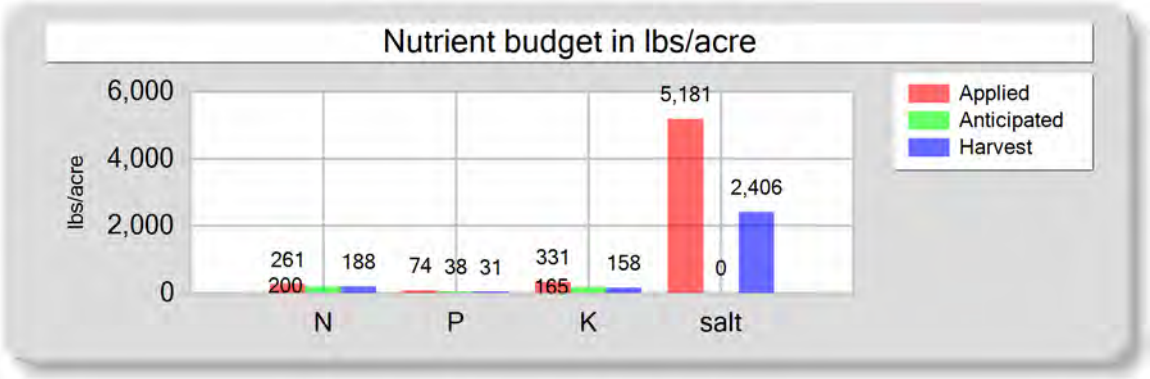
Fresh water applied	
36,540,000.00	*gallons*
1,345.64	*acre-inches*
5.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Lower Jones - 06/22/2023: Corn, silage

Field name: Lower Jones Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	221.95	73.84	331.34	4,470.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	710.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.95	73.84	331.34	5,180.67
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	188.01	30.75	158.28	2,406.19
Nutrient balance	72.93	43.09	173.07	2,774.48
Applied to removed ratio	1.39	2.40	2.09	2.15

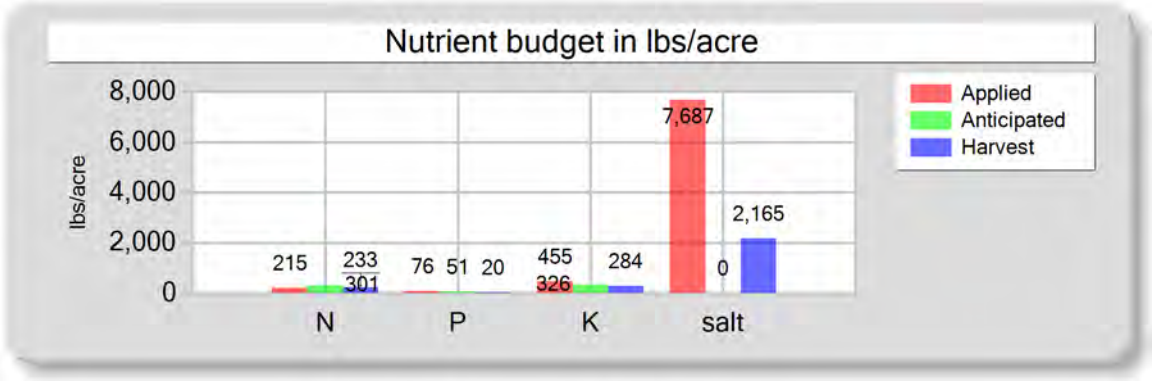
Fresh water applied	
222,120,000.00	*gallons*
8,179.92	*acre-inches*
31.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Prune Ranch - 02/01/2014: Almond, in shell

Field name: Prune Ranch Crop: Almond, in shell Plant date: 02/01/2014



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	131.08	76.07	454.80	6,524.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,162.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	215.48	76.07	454.80	7,686.73
Anticipated crop nutrient removal	301.00	51.10	325.50	0.00
Actual crop nutrient removal	233.33	19.87	284.00	2,164.80
Nutrient balance	-17.86	56.20	170.80	5,521.93
Applied to removed ratio	0.92	3.83	1.60	3.55

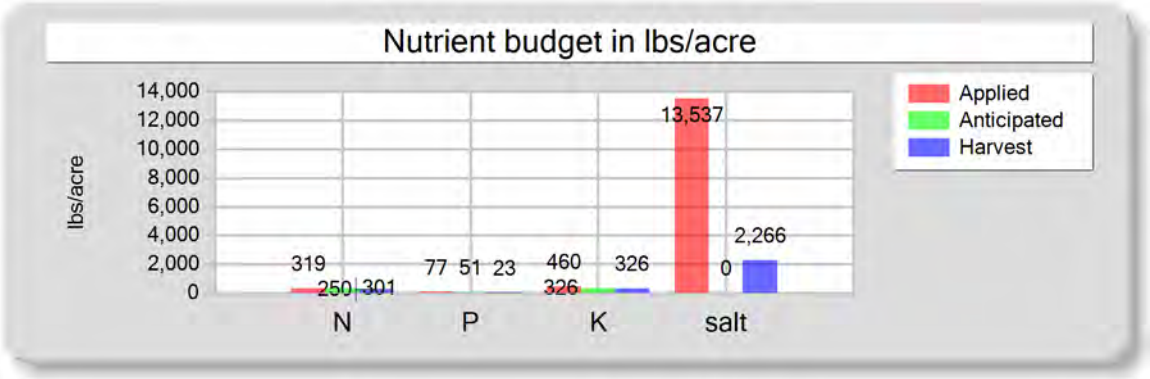
Fresh water applied	
47,875,128.00	*gallons*
1,763.08	*acre-inches*
27.99	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

River Ranch - 02/01/2007: Almond, in shell

Field name: River Ranch Crop: Almond, in shell Plant date: 02/01/2007



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	70.40	0.00	0.00	0.00
Dry manure	132.44	76.85	459.52	6,591.82
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	102.58	0.00	0.00	6,944.76
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	319.42	76.85	459.52	13,536.58
Anticipated crop nutrient removal	301.00	51.10	325.50	0.00
Actual crop nutrient removal	249.99	22.63	326.30	2,265.78
Nutrient balance	69.43	54.22	133.22	11,270.80
Applied to removed ratio	1.28	3.40	1.41	5.97

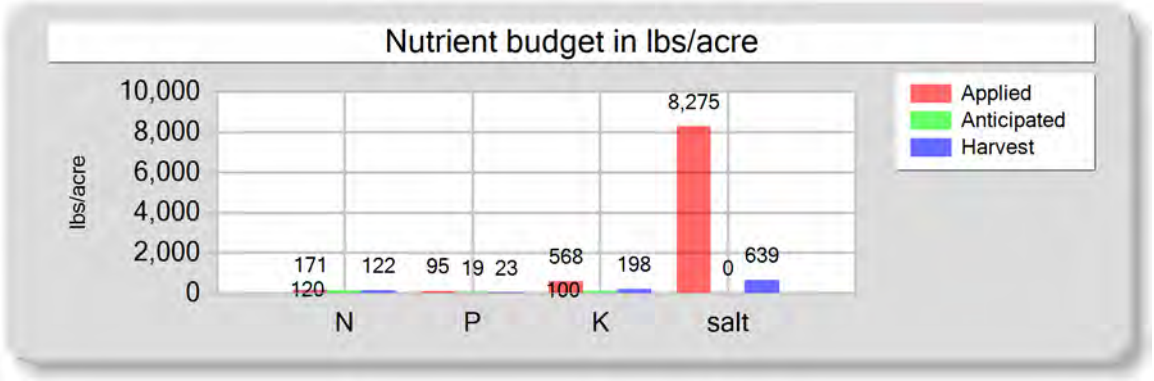
Fresh water applied	
121,395,228.00	*gallons*
4,470.57	*acre-inches*
28.12	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Willows Ranch - 11/30/2022: Oats, silage-soft dough

Field name: Willows Ranch Crop: Oats, silage-soft dough Plant date: 11/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.81	95.06	568.36	8,153.09
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	121.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.81	95.06	568.36	8,274.92
Anticipated crop nutrient removal	120.00	19.20	99.60	0.00
Actual crop nutrient removal	121.76	23.00	198.02	638.51
Nutrient balance	49.04	72.06	370.34	7,636.41
Applied to removed ratio	1.40	4.13	2.87	12.96

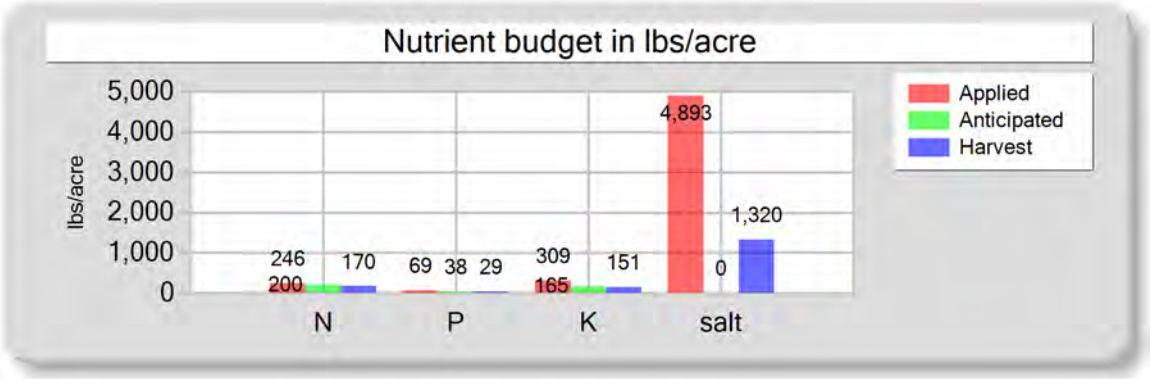
Fresh water applied	
57,960,000.00	*gallons*
2,134.47	*acre-inches*
5.38	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Willows Ranch - 06/24/2023: Corn, silage

Field name: Willows Ranch Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	32.00	0.00	0.00	0.00
Dry manure	206.65	68.75	308.50	4,162.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	730.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.65	68.75	308.50	4,892.54
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	170.26	29.26	150.59	1,320.23
Nutrient balance	75.39	39.50	157.91	3,572.31
Applied to removed ratio	1.44	2.35	2.05	3.71

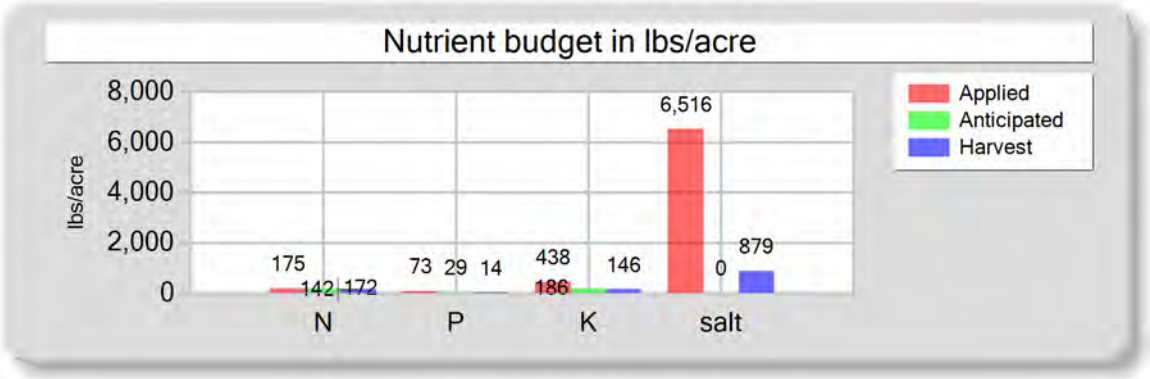
Fresh water applied	
347,400,000.00	*gallons*
12,793.56	*acre-inches*
32.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Woodland Ranch - 02/01/2020: Almond, in shell

Field name: Woodland Ranch Crop: Almond, in shell Plant date: 02/01/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	126.16	73.21	437.75	6,279.46
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	236.80
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	175.36	73.21	437.75	6,516.26
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	142.33	13.62	145.72	879.21
Nutrient balance	33.03	59.59	292.02	5,637.05
Applied to removed ratio	1.23	5.37	3.00	7.41

Fresh water applied	
15,059,904.00	*gallons*
554.61	*acre-inches*
30.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 21.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	7,660.00	45,800.00							65.70
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,100.00	7,020.00	31,500.00							42.50
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 05/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 83.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,700.00	7,590.00	15,000.00							17.40
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,500.00	7,560.00	48,600.00							42.50
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	3,530.00	8,160.00							55.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 10/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	605.00	322.00	0.00	0.00	140.00	850.00								6,570.00	3,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 02/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,150.00	501.00	0.00	0.00	180.00	915.00								7,190.00	5,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,240.00	577.00	0.00	0.00	68.70	1,100.00	202.00	120.00	246.00	4,880.00	0.00	321.00	485.00	8,330.00	4,600
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,140.00	619.00	0.00	0.00	65.70	1,230.00								8,950.00	6,380
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	960.00	515.00	0.00	0.00	106.00	952.00								8,230.00	4,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 08/17/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,940.00	495
DL	0.50	0.50	0.50								1.00	10

Canal

Sample description: Canal

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	1.00	0.50	0.10								10.00	10

IW #1 Shackleford

IW #1 Shackleford

Sample description: IW #1 Shackleford

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	45.00	0.00	0.00								1,150.00	700
DL	0.50	0.50	0.50								1.00	10

IW #3 Woodland

IW #3 Woodland

IW #3 Woodland

Sample description: IW #3 Woodland

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.00	0.00	0.00								556.00	285
DL	0.50	0.50	0.50								1.00	10

IW #4 River Road.

IW #4 River Road

Sample description: IW #4 River Road

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.10	0.00	0.00								1,910.00	1,090
DL	0.50	0.50	0.50								1.00	10

IW #5 Prune

IW #5 Prune

Sample description: IW #5 Prune

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	18.10	0.00	0.00								1,370.00	870
DL	0.50	0.50	0.50								1.00	10

IW #6 Greenwood South

IW #6 Greenwood South

IW #6 Greenwood South

Sample description: IW #6 Greenwood South

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.90	0.00	0.00								1,260.00	705
DL	0.50	0.50	0.50								1.00	10

IW #7 Greenwood NW

IW #7 Greenwood NW

Sample description: IW #7 Greenwood NW

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.04	0.00	0.00								9,480.00	5,940
DL	0.50	0.50	0.50								1.00	10

IW #8 Greenwood NE

IW #8 Greenwood NE

Sample description: IW #8 Greenwood NE

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.40	0.00	0.00								208.00	1,200
DL	0.50	0.50	0.50								1.00	10

IW#2 Gates

IW#2 Gates

IW #2 Gates

Sample description: IW #2 Gates

Sample date: 12/16/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	43.50	0.00	0.00								1,430.00	850
DL	0.50	0.50	0.50								1.00	10

MID Lateral 3

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.70	37
DL	1.00	0.50	0.50								10.00	10

River

River

River

Sample description: River

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								163.00	100
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

1

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

2

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			79.00				
DL			4.00				

3

3

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			98.00				
DL			4.00				

4

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			98.00				
DL			4.00				

5

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			86.00				
DL			4.00				

6

6

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			90.00				
DL			4.00				

7

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			110.00				
DL			4.00				

Blue Gum

Soil

Sample and source description: Soil

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 11/15/2022: Oats, silage-soft dough

1 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,430.00	15,300.00		11.60
DL	100.00	100.00	100.00		0.01

1 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,230.00	520.00	3,110.00		6.80
DL	100.00	100.00	100.00		0.01

2 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,000.00	1,790.00	13,100.00		10.40
DL	100.00	100.00	100.00		0.01

2 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,940.00	460.00	2,530.00		6.80
DL	100.00	100.00	100.00		0.01

3 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	1,720.00	23,900.00		14.80
DL	100.00	100.00	100.00		0.01

3 - 06/28/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 08/11/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,170.00	1,410.00	16,200.00		12.90
DL	100.00	100.00	100.00		0.01

4 - 11/15/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,500.00	1,850.00	12,600.00		13.20
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,920.00	1,960.00	13,700.00		10.10
DL	100.00	100.00	100.00		0.01

4 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,150.00	480.00	2,660.00		7.10
DL	100.00	100.00	100.00		0.01

4 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,010.00	430.00	2,520.00		5.70
DL	100.00	100.00	100.00		0.01

5 - 11/16/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,690.00	1,710.00	15,600.00		11.90
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,240.00	1,570.00	14,000.00		10.60
DL	100.00	100.00	100.00		0.01

5 - 06/03/2023: Corn, silage

5 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,960.00	460.00	2,600.00		5.70
DL	100.00	100.00	100.00		0.01

6 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,520.00	1,360.00	10,400.00		11.20
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,880.00	1,580.00	12,900.00		10.80
DL	100.00	100.00	100.00		0.01

6 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,050.00	1,550.00	13,500.00		12.90
DL	100.00	100.00	100.00		0.01

6 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,970.00	440.00	2,470.00		7.50
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,810.00	460.00	2,510.00		5.90
DL	100.00	100.00	100.00		0.01

6 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,900.00	440.00	2,350.00		6.40
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	490.00	2,990.00		6.60
DL	100.00	100.00	100.00		0.01

7 - 11/16/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,190.00	1,450.00	12,300.00		13.00
DL	100.00	100.00	100.00		0.01

7 - 06/03/2023: Corn, silage

7 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,870.00	480.00	2,780.00		5.90
DL	100.00	100.00	100.00		0.01

Bacon Island - 11/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,410.00	980.00	8,720.00		11.60
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,690.00	970.00	2,920.00		12.60
DL	100.00	100.00	100.00		0.01

Bacon Island - 11/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,340.00	910.00	7,730.00		11.40
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,670.00	910.00	7,530.00		7.20
DL	100.00	100.00	100.00		0.01

Bacon Island - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	670.00	3,690.00		14.80
DL	100.00	100.00	100.00		0.01

Bacon Island - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	690.00	3,740.00		21.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	680.00	3,340.00		19.10
DL	100.00	100.00	100.00		0.01

Blue Gum - 11/17/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage
Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,080.00	1,740.00	12,900.00		12.40
DL	100.00	100.00	100.00		0.01

Blue Gum - 11/17/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,800.00	1,350.00	11,100.00		13.20
DL	100.00	100.00	100.00		0.01

Blue Gum - 06/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,360.00	500.00	2,960.00		6.40
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,250.00	450.00	2,580.00		6.30
DL	100.00	100.00	100.00		0.01

Greenwood - 02/01/2017: Almond, in shell

Greenwood - 02/01/2017: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 6.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	1,890.00	21,400.00		15.20
DL	100.00	100.00	100.00		0.01

Lower Jones - 11/29/2022: Oats, silage-soft dough

F

Sample and source description: F

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,830.00	980.00	8,570.00		11.10
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,140.00	860.00	7,710.00		9.90
DL	100.00	100.00	100.00		0.01

Lower Jones - 06/22/2023: Corn, silage

Lower Jones - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,440.00	600.00	3,450.00		19.10
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,250.00	640.00	3,340.00		13.20
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,730.00	610.00	3,140.00		15.60
DL	100.00	100.00	100.00		0.01

Prune Ranch - 02/01/2014: Almond, in shell

Prune Ranch - 02/01/2014: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	1,490.00	21,300.00		18.00
DL	100.00	100.00	100.00		0.01

River Ranch - 02/01/2007: Almond, in shell

Almond

Sample and source description: Almond

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,000.00	1,720.00	24,800.00		18.80
DL	100.00	100.00	100.00		0.01

Willows Ranch - 11/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	950.00	8,230.00		12.10
DL	100.00	100.00	100.00		0.01

Willows Ranch - 11/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,870.00	920.00	7,920.00		11.30
DL	100.00	100.00	100.00		0.01

Willows Ranch - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,150.00	560.00	2,950.00		14.70
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,550.00	610.00	3,140.00		9.30
DL	100.00	100.00	100.00		0.01

Willows Ranch - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,220.00	550.00	3,000.00		13.30
DL	100.00	100.00	100.00		0.01

Woodland Ranch - 02/01/2020: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/31/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	2,010.00	21,500.00		14.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

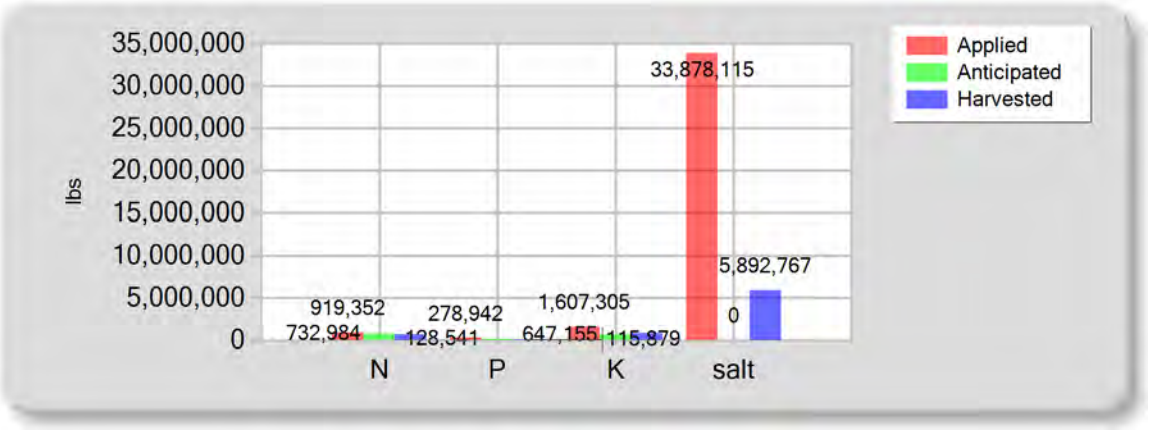
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

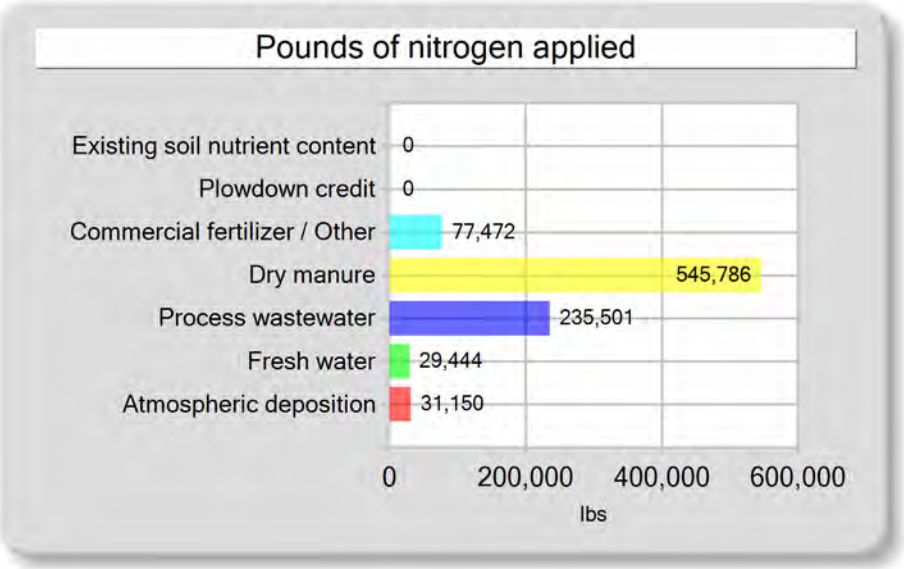
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

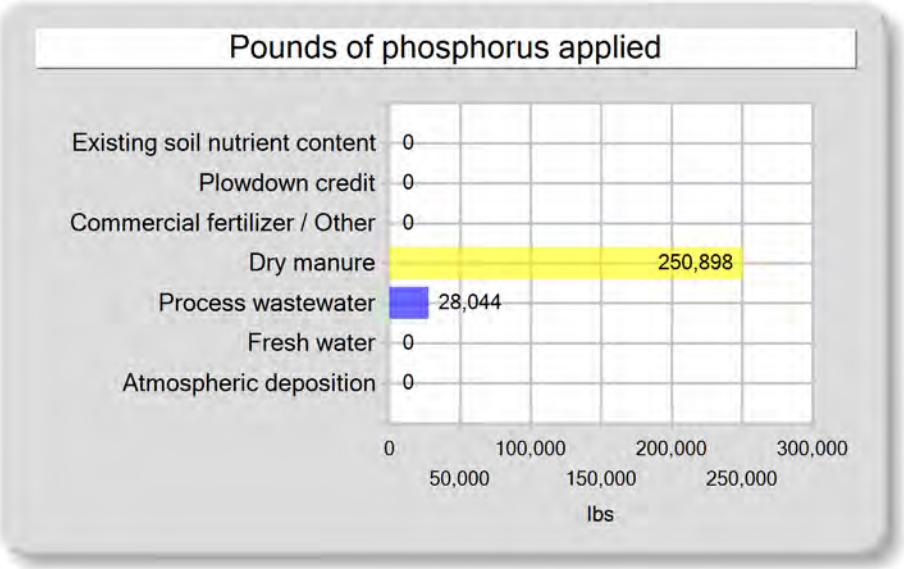
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	77,472.00	0.00	0.00	0.00
Dry manure	545,785.76	250,898.16	1,368,192.61	19,288,193.88
Process wastewater	235,500.66	28,043.75	239,112.50	1,198,851.94
Fresh water	29,443.60	0.00	0.00	13,391,069.17
Atmospheric deposition	31,150.00	0.00	0.00	0.00
Total nutrients applied	919,352.02	278,941.90	1,607,305.11	33,878,114.99
Anticipated crop nutrient removal	732,984.00	128,541.30	647,154.60	0.00
Actual crop nutrient removal	714,953.06	115,879.36	798,511.62	5,892,766.85
Nutrient balance	204,398.96	163,062.54	808,793.49	27,985,348.15
Applied to removed ratio	1.29	2.41	2.01	5.75

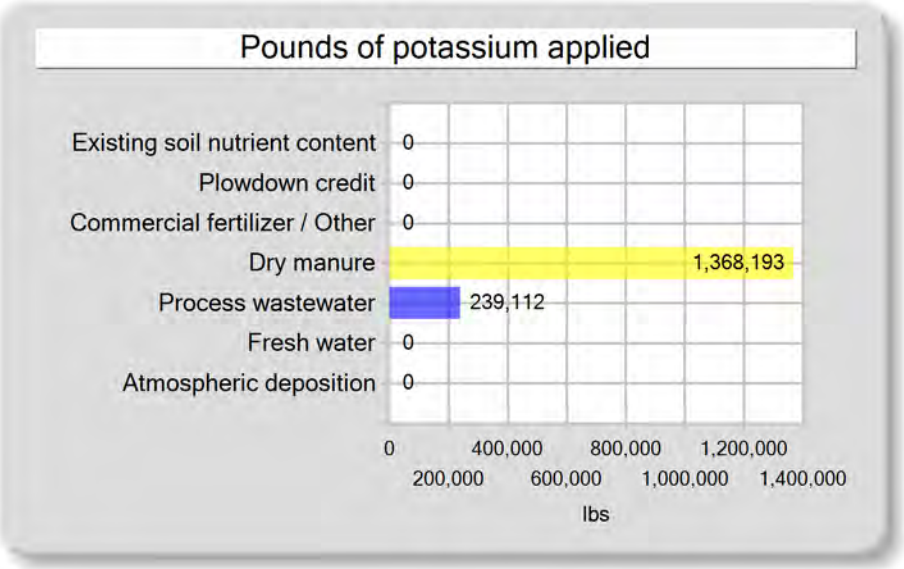
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

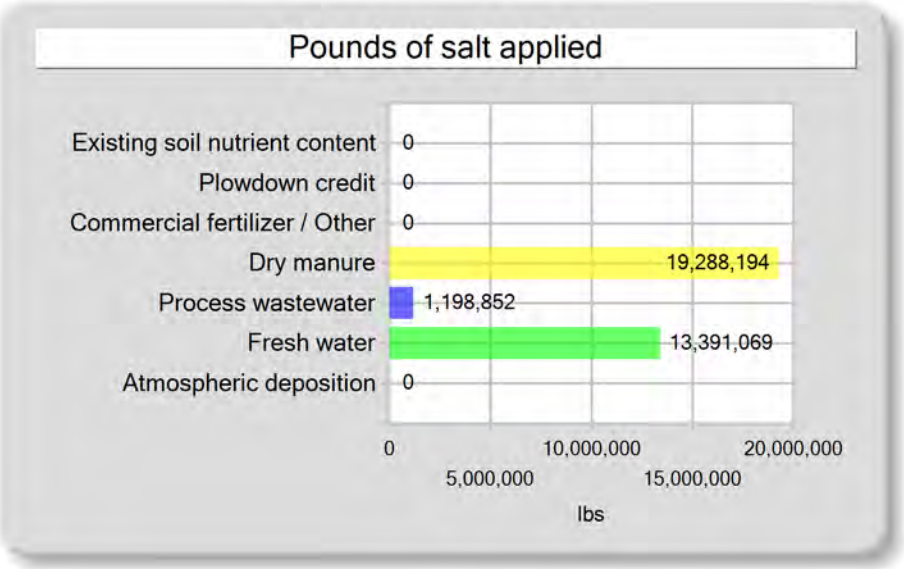


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mark Meirinho (Jun 25, 2024 17:55 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mark Meirinho	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Meirinho West, Inc.
5418 Woodland Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/11/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Milk Barn	T3223037A	Electrical Conductivity (EC)	EPA 120.1 and	0.453	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	5.72	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	290	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.0	93.4 (T3223024A)	95.2 (T3223024A)	1.90
Electrical Conductivity (EC)	99.9			0.230
Nitrate Nitrogen (NO3-N)	100	103 (W3222002A)	99.8 (W3222002A)	2.96

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Grower

Consultant:
F & R Ag Services
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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/11/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - Shop	T3223037B	Electrical Conductivity (EC)	EPA 120.1 and	0.417	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.21	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	267	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.0	93.4 (T3223024A)	95.2 (T3223024A)	1.90
Electrical Conductivity (EC)	99.9			0.230
Nitrate Nitrogen (NO3-N)	100	103 (W3222002A)	99.8 (W3222002A)	2.96

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PURCHASE ORDER:
RECEIVED DATE: 8/11/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - LMT Milk Barn	T3223037C	Electrical Conductivity (EC)	EPA 120.1 and	1.74	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	47.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1110	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.0	93.4 (T3223024A)	95.2 (T3223024A)	1.90
Electrical Conductivity (EC)	99.9			0.230
Nitrate Nitrogen (NO3-N)	100	103 (W3222002A)	99.8 (W3222002A)	2.96

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/11/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - LMT House	T3223037D	Electrical Conductivity (EC)	EPA 120.1 and	1.80	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	48.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1150	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.0	93.4 (T3223024A)	95.2 (T3223024A)	1.90
Electrical Conductivity (EC)	99.9			1.01
Nitrate Nitrogen (NO3-N)	100	103 (W3222002A)	99.8 (W3222002A)	2.96

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/11/23 3:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 8/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 23.6 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW - LMT Heifers	T3223037E	Electrical Conductivity (EC)	EPA 120.1 and	2.46	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	33.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1580	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.0	93.4 (T3223024A)	95.2 (T3223024A)	1.90
Electrical Conductivity (EC)	99.9			1.01
Nitrate Nitrogen (NO3-N)	100	103 (W3222002A)	99.8 (W3222002A)	2.96

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DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MEIRINHO WEST
Mailing address:
(not facility address) MODESTO CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Sean Locke
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/11/23	11:00	DW- MILK BARN	DOMESTIC	23.4		X				
2		11:05	DW- SHOP		23.8		X				
3		11:10	DW-LMT MILK BARN		23.8		X				
4		11:15	DW-LMT HOUSE		23.4		X				
5		11:20	DW-LMT HEIFERS		23.6		X				
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	8-11-23 @ 15:40			T3223037

Required to be submitted to any agency? (YES / NO)

2023 Meirinho West Annual Report

Final Audit Report 2024-06-26

Created:	2024-06-25
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAANCsIqW8flqbt__GC7VrOp4ueHz-dy0dc

"2023 Meirinho West Annual Report" History

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